SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME:Data Warehousing and Data Mining   COURSE CODE: CSA1610

COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets

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QUESTIONS: 05   **Total Marks:50**

Annexure A

Constraint-Based Problem Solving

Criteria	Problem Understanding *(2.5)*	Constraint Analysis *(2.5)*	Solution Development *(2.5)*	Application of Concepts *(1.5)*	Justification *(1)*	*Total(10)*
Weightage	*25%*	25%	25%	15%	10%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I G.Hima Bindu(192511377) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G.Hima Bindu

RUBRICS FOR CALCULATION:

Constraint-Based Problem Solving Assessment Rubric

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Annexure A

Q1. Dataset: Hospital Patient Records

PID	Symptoms Observed
P1	Fever, Cough
P2	Fever, Headache
P3	Cough, Headache
P4	Fever, Cough, Headache
P5	Fever, Fatigue

Question:
Apply association rule mining with minimum support = 50% and confidence = 70%, under the constraint that all generated rules must include **"Fever"**. Identify the valid association rules and justify why they satisfy the given constraints.

Q2. Dataset: Online Learning Platform

Student ID	Courses Enrolled
S1	Python, SQL
S2	Python, Data Science
S3	SQL, Machine Learning
S4	Python, SQL, Data Science
S5	Python, Machine Learning

Question:
Using the dataset, generate frequent itemsets with minimum support = 40%, applying the constraint that each itemset must contain **at least two courses**. Explain how the constraint helps in pruning candidate itemsets during mining.

Q3. Dataset: Library Borrowing Records

Transaction ID	Books Borrowed
T1	Data Mining, Python Programming
T2	Data Mining, Database Systems
T3	Python Programming, Database Systems
T4	Data Mining, Python Programming, Database Systems
T5	Data Mining, Artificial Intelligence

Question:
Apply the Apriori algorithm with the constraint that the **maximum itemset size is 3**. Identify the frequent itemsets and explain how the constraint reduces computational complexity during candidate generation.

Q4. Dataset: University Course Hierarchy

Student ID	Courses Taken
S1	Engineering, Computer Science, Machine Learning
S2	Engineering, Information Technology
S3	Engineering, Computer Science
S4	Engineering, Computer Science, Machine Learning
S5	Engineering, Information Technology

Question:
Apply multilevel association rule mining with the constraint that rules should be generated only at the **"Computer Science"** level. Explain how the course hierarchy affects the discovery of association rules.

Q5. Dataset: Insurance Customer Records

CID	Age Group	Income Level	Policy Purchased
C1	Young	High	Health Insurance
C2	Middle	Medium	Vehicle Insurance
C3	Young	Low	Travel Insurance
C4	Senior	High	Health Insurance
C5	Young	Medium	Vehicle Insurance

Question:
Apply multidimensional association rule mining with the constraint **Age Group = "Young"**. Identify meaningful patterns involving income level and policy type, and explain how these patterns can support targeted marketing strategies.

Q1. Hospital Patient Records

Given Dataset

There are 5 patient records:

Patient	**Symptoms**
P1	Fever, Cough
P2	Fever, Headache
P3	Cough, Headache
P4	Fever, Cough, Headache
P5	Fever, Fatigue

Minimum Support = **50%**
Minimum Confidence = **70%**
Constraint = Every generated rule must contain **Fever**.

Step 1: Find Support of Individual Symptoms

The formula for support is:

[Support(X)=\frac{\text{Number of transactions containing X}}{\text{Total transactions}}\times100]

Total transactions = 5.

For minimum support of 50%:

[5\times0.50=2.5]

Therefore, an itemset must occur in at least **3 transactions**.

Symptom	**Count**	**Support**
Fever	4	80%
Cough	3	60%
Headache	3	60%
Fatigue	1	20%

Thus, the frequent 1-itemsets are:

$$
\{Fever\},\{Cough\},\{Headache\}
$$

Fatigue is removed because its support is only 20%.

Step 2: Generate Itemsets Containing Fever

Because of the constraint, we only consider itemsets that contain Fever.

{Fever, Cough}

This occurs in P1 and P4.

$$Support(Fever,Cough)=\frac{2}{5}\times100=40\%$$

Since 40% < 50%, it is not frequent.

{Fever, Headache}

This occurs in P2 and P4.

$$Support(Fever,Headache)=\frac{2}{5}\times100=40\%$$

It is also not frequent.

{Fever, Fatigue}

This occurs only in P5.

$$Support(Fever,Fatigue)=\frac{1}{5}\times100=20\%$$

So it is not frequent.

Step 3: Check Confidence

Although the above pairs fail the support constraint, their confidence can also be checked.

For:

$$Fever\rightarrow Cough$$

$$Confidence=\frac{Support(Fever,Cough)}{Support(Fever)}$$

Using transaction counts:

[=\frac{2}{4}\times100=50%]

This is below 70%.

For:

[Cough\rightarrow Fever]

[=\frac{2}{3}\times100=66.67%]

This is also below 70%.

Similarly:

[Confidence(Fever\rightarrow Headache)=\frac{2}{4}\times100=50%]

[Confidence(Headache\rightarrow Fever)=\frac{2}{3}\times100=66.67%]

Both fail the 70% confidence requirement.

Final Result

There are **no valid association rules** satisfying all three conditions simultaneously:

1. Minimum support = 50%
2. Minimum confidence = 70%
3. Rule must contain Fever

Therefore,

[\boxed{\text{Valid association rules = None}}]

How the Constraint Helps

The Fever constraint reduces the search space because the algorithm does not need to generate rules involving only Cough, Headache or Fatigue. It concentrates only on patterns containing Fever. However, after support and confidence checking, no rule remains valid.

Q2. Online Learning Platform

Given Dataset

Student	Courses
S1	Python, SQL
S2	Python, Data Science
S3	SQL, Machine Learning
S4	Python, SQL, Data Science
S5	Python, Machine Learning

Minimum Support = **40%**

Constraint = Every final itemset must contain **at least two courses**.

There are 5 transactions.

$$[Minimum\ count=5\times0.40=2]$$

Therefore, an itemset must occur at least **2 times**.

Step 1: 1-Itemsets

Course	Count	Support
Python	4	80%
SQL	2	40%
Data Science	2	40%
Machine Learning	2	40%

All four courses satisfy the support threshold.

However, the constraint says that the required itemset must contain **at least two courses**. Therefore, single-course itemsets cannot be final answers.

Step 2: Generate 2-Itemsets

Python, SQL

Present in S1 and S4.

$$[Support=\frac{2}{5}\times100=40\%]$$

Frequent.

Python, Data Science

Present in S2 and S4.

$$[Support=\frac{2}{5}\times100=40\%]$$

Frequent.

Python, Machine Learning

Present only in S5.

$$[Support=20\%]$$

Not frequent.

SQL, Data Science

Present only in S4.

[Support=20%]

Not frequent.

SQL, Machine Learning

Present only in S3.

[Support=20%]

Not frequent.

Data Science, Machine Learning

No student has both courses.

[Support=0%]

Not frequent.

Therefore, the frequent 2-itemsets are:

[\boxed{{Python,SQL}}]

and

[\boxed{{Python,Data\ Science}}]

Both have 40% support.

Step 3: 3-Itemsets

The main possible 3-itemset is:

[{Python,SQL,Data\ Science}]

It occurs only in S4.

[Support=\frac{1}{5}\times100=20%]

Thus, it is not frequent.

Other 3-itemsets also fail the support requirement.

Final Frequent Itemsets

Itemset	Support
{Python, SQL}	40%
{Python, Data Science}	40%

Role of the Constraint

The constraint **minimum itemset size = 2** prevents single-course itemsets from being considered as final results.For example, {Python} has 80% support, but it is not useful for this particular query because the user wants combinations of at least two courses.

The constraint therefore:

- saves computation and memory,
- focuses the mining process on course combinations.
- Thus, constraint-based mining is more efficient than blindly generating every possible itemset.

Q3. Library Borrowing Records

Given Dataset

Transaction	Books
T1	Data Mining, Python Programming
T2	Data Mining, Database Systems
T3	Python Programming, Database Systems
T4	Data Mining, Python Programming, Database Systems
T5	Data Mining, Artificial Intelligence

The maximum itemset size is **3**.

Important Note

The question does not specify minimum support. To determine which itemsets are **frequent**, we use **40% minimum support**, which gives a minimum count of 2 out of 5 transactions.

[Minimum\ count=5\times40%=2]

Step 1: Count 1-Itemsets

Book	Count	Support
Data Mining	4	80%
Python Programming	3	60%
Database Systems	3	60%
Artificial Intelligence	1	20%

Artificial Intelligence is pruned because:

[20%<40%]

Frequent 1-itemsets are therefore:

[{Data\ Mining},{Python\ Programming},{Database\ Systems}]

Step 2: Candidate 2-Itemsets

Data Mining + Python Programming

Occurs in T1 and T4.

[Support=\frac{2}{5}\times100=40%]

Frequent.

Data Mining + Database Systems

Occurs in T2 and T4.

[Support=\frac{2}{5}\times100=40%]

Frequent.

Python Programming + Database Systems

Occurs in T3 and T4.

[Support=\frac{2}{5}\times100=40%]

Frequent.

So the frequent 2-itemsets are:

[\boxed{{Data\ Mining,Python\ Programming}}]

[\boxed{{Data\ Mining,Database\ Systems}}]

[\boxed{{Python\ Programming,Database\ Systems}}]

Step 3: Candidate 3-Itemset

The possible 3-itemset is:

[{Data\ Mining,Python\ Programming,Database\ Systems}]

It occurs only in T4.

[Support=\frac{1}{5}\times100=20%]

Since 20% < 40%, it is not frequent.

Therefore, there are **no frequent 3-itemsets**.

Final Frequent Itemsets

1-itemsets:

- Data Mining – 80%
- Python Programming – 60%
- Database Systems – 60%

2-itemsets:

- Data Mining + Python Programming – 40%
- Data Mining + Database Systems – 40%
- Python Programming + Database Systems – 40%

3-itemsets:

- None

Effect of Maximum Size = 3

Apriori normally tries to create larger itemsets as long as the support condition allows them.
Here, the constraint states:

[|Itemset|\leq3]

Therefore, the algorithm stops after checking 3-itemsets.

It does not generate:

- 4-itemsets
- 5-itemsets
- larger combinations

This greatly reduces the number of possible candidates.

For example, without the constraint, the algorithm could continue joining frequent sets to search for larger combinations. With the maximum-size constraint, the search becomes:

[L_1\rightarrow L_2\rightarrow L_3]

and then stops.

Thus, the constraint reduces candidate generation, database scans, processing time and memory requirement.

Q4. University Course Hierarchy

Given Dataset

Student	Courses
S1	Engineering, Computer Science, Machine Learning
S2	Engineering, Information Technology
S3	Engineering, Computer Science
S4	Engineering, Computer Science, Machine Learning
S5	Engineering, Information Technology

Constraint:

Rules must be generated only at the Computer Science level.

Course Hierarchy

The hierarchy can be viewed as:

Engineering

│

├── Computer Science

│ └── Machine Learning

│

└── Information Technology

This means Engineering is a higher-level category, Computer Science is a lower level, and Machine Learning is a more specific concept under Computer Science.

Step 1: Identify Computer Science Students

Computer Science appears in:

- S1
- S3
- S4

Therefore:

$$Support(Computer\ Science)=\frac{3}{5}\times100=60\%$$

Machine Learning appears in:

- S1
- S4

Therefore:

$$Support(Machine\ Learning)=\frac{2}{5}\times100=40\%$$

Computer Science and Machine Learning occur together in S1 and S4:

$$Support(CS,ML)=\frac{2}{5}\times100=40\%$$

Step 2: Generate Rules at the Required Level

The rule:

$$Machine\ Learning\rightarrow Computer\ Science$$

has confidence:

$$Confidence=\frac{Support(Machine\ Learning,CS)}{Support(Machine\ Learning)}$$

$$=\frac{2}{2}\times100=100\%$$

Thus:

$$\boxed{Machine\ Learning\rightarrow Computer\ Science}$$

has:

- **Support = 40%**
- **Confidence = 100%**

The reverse direction is:

$$Computer\ Science\rightarrow Machine\ Learning$$

$$Confidence=\frac{2}{3}\times100=66.67\%$$

So the reverse rule is weaker.

Why the Hierarchy is Important

Multilevel association mining uses the relationship between general and specific concepts.

At the higher level:

[Engineering]

has very high occurrence because every student belongs to Engineering. However, such a rule is too general and does not provide much useful information.

At the Computer Science level, the relationship becomes more meaningful:

[Machine\ Learning\right arrow Computer\ Science]

The hierarchy allows the mining system to move from broad concepts to more specific concepts.

The constraint also avoids generating rules involving Information Technology, because those rules are outside the requested Computer Science level.

Final Result

The important Computer Science-level pattern is:

[\boxed{Machine\ Learning\rightarrow Computer\ Science}]

with **40% support and 100% confidence**.

The course hierarchy makes it possible to discover relationships at different abstraction levels and allows the user to focus only on the required level.

Q5. Insurance Customer Records

Given Dataset

CID	Age Group	Income	Policy
C1	Young	High	Health Insurance
C2	Middle	Medium	Vehicle Insurance
C3	Young	Low	Travel Insurance
C4	Senior	High	Health Insurance
C5	Young	Medium	Vehicle Insurance

Constraint:

[Age\ Group=Young]

This means we first filter the dataset and consider only Young customers.

Step 1: Apply Age Constraint

The selected records are:

Customer	Income	Policy
C1	High	Health Insurance
C3	Low	Travel Insurance
C5	Medium	Vehicle Insurance

Total Young customers = **3**.

The mining task is now reduced from 5 records to only 3 records.

Step 2: Discover Income Patterns

Among Young customers:

- High income → 1
- Medium income → 1
- Low income → 1

Therefore each income group has:

$$\frac{1}{3}\times100=33.33\%$$

support within the Young subset.

Step 3: Discover Policy Patterns

For Young customers:

- High income → Health Insurance
- Medium income → Vehicle Insurance
- Low income → Travel Insurance

Therefore, the following multidimensional relationships are observed.

Pattern 1

$$Young+High\ Income\rightarrow Health\ Insurance$$

Support in complete dataset:

$$\frac{1}{5}\times100=20\%$$

Confidence among High-income Young customers:

$$\frac{1}{1}\times100=100\%$$

Pattern 2

$$Young+Medium\ Income\rightarrow Vehicle\ Insurance$$

Support:

$$\frac{1}{5}\times100=20\%$$

Confidence within the corresponding Young + Medium segment:

$$100\%$$

Pattern 3

$$Young+Low\ Income\rightarrow Travel\ Insurance$$

Support:

$$\frac{1}{5}\times100=20\%$$

Confidence:

$$100\%$$

These confidence values are 100% in this small dataset because each filtered segment contains one customer and that customer has the indicated policy. Therefore, they should be interpreted as sample patterns, not as strong general conclusions for a larger population.

Step 4: Interpretation for Marketing

Young + High Income

The customer represented by C1 purchased Health Insurance.

This suggests that premium health insurance products, family health plans or plans with additional benefits could be promoted toward similar high-income young customers.

Young + Medium Income

C5 purchased Vehicle Insurance.

The company can market vehicle policies with affordable premiums, installment options and basic-to-medium coverage to this segment.

Young + Low Income

C3 purchased Travel Insurance.

The company can promote low-cost travel insurance, student-oriented plans or short-term travel policies to similar customers.

How Multidimensional Mining Helps

Ordinary association mining may find only one-dimensional relationships such as:

Young → Insurance

This does not provide enough detail.

Multidimensional association mining combines multiple attributes:

[Age + Income + Policy]

So the company receives more useful patterns such as:

[Young+High\ Income\rightarrow Health\ Insurance]

This supports **customer segmentation and personalized marketing**.

Targeted Marketing Table

Customer Segment	Observed Policy	Suitable Marketing Approach
Young + High Income	Health Insurance	Premium health plans
Young + Medium Income	Vehicle Insurance	Affordable vehicle plans
Young + Low Income	Travel Insurance	Low-cost travel plans

Final Answer

The constraint Age Group = Young reduces the search space and allows the company to concentrate on one important customer segment. The observed patterns show different policy preferences across income levels.

Therefore, multidimensional association mining can help an insurance company with:

- targeted advertisements,
- personalized policy recommendations,
- customer segmentation,
- customized pricing and offers,
- better allocation of marketing resources.

The major patterns identified are:

[\boxed{Young+High\ Income\rightarrow Health\ Insurance}]

[\boxed{Young+Medium\ Income\rightarrow Vehicle\ Insurance}]

[\boxed{Young+Low\ Income\rightarrow Travel\ Insurance}]

These patterns demonstrate how constraint-based multidimensional mining can convert raw customer records into practical marketing decisions.